MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG5DSCSTA300 – APPLIED REGRESSION ANALYSIS

Duration: 2 Hours **Maximum: 70 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer any ten questions. Each question carries **3 marks**.

1. Define simple linear regression. [U] [1]
2. Mention any three applications of regression analysis. [U] [1]
3. Write the general form of a multiple linear regression model. [U][2]
4. What is the role of $(X'X)^{-1}$ in OLS estimation? [An][2]
5. State the formula for R^2 and specify its range. [A][2]
6. What is meant by serial correlation in regression analysis? [A][2]
7. Define the Generalized Least Squares (GLS) method. [An][3]
8. What is nonparametric regression. [U][3]
9. Define Poisson regression. [U][4]
10. What is meant by classification in logistic regression? [A][4]
11. What is meant by a fixed effect in a regression model? [U][5]
12. State two causes of multicollinearity. [U][5]

Part B

Short Essay questions

Answer any four questions. Each question carries **7 marks.**

13. Explain the assumptions of the simple linear regression model. [K][1]
14. Explain the OLS method for estimating coefficients in multiple linear regression. [A][2]
15. Explain the need for model selection in multiple regression. [A][2]
16. Explain Weighted Least Squares and discuss its purpose. [A][3]
17. Explain the role of the reference category in multinomial logistic regression. [An][4]
18. Explain mixed effect models and their components. [A][5]

Part C

Essay questions

Answer any one question. Each question carries **12 marks**.

19. Explain the role of regression analysis in statistical modeling and decision-making. [A][1]
20. Explain forward selection, backward elimination, and stepwise selection procedures in detail. [A][3]